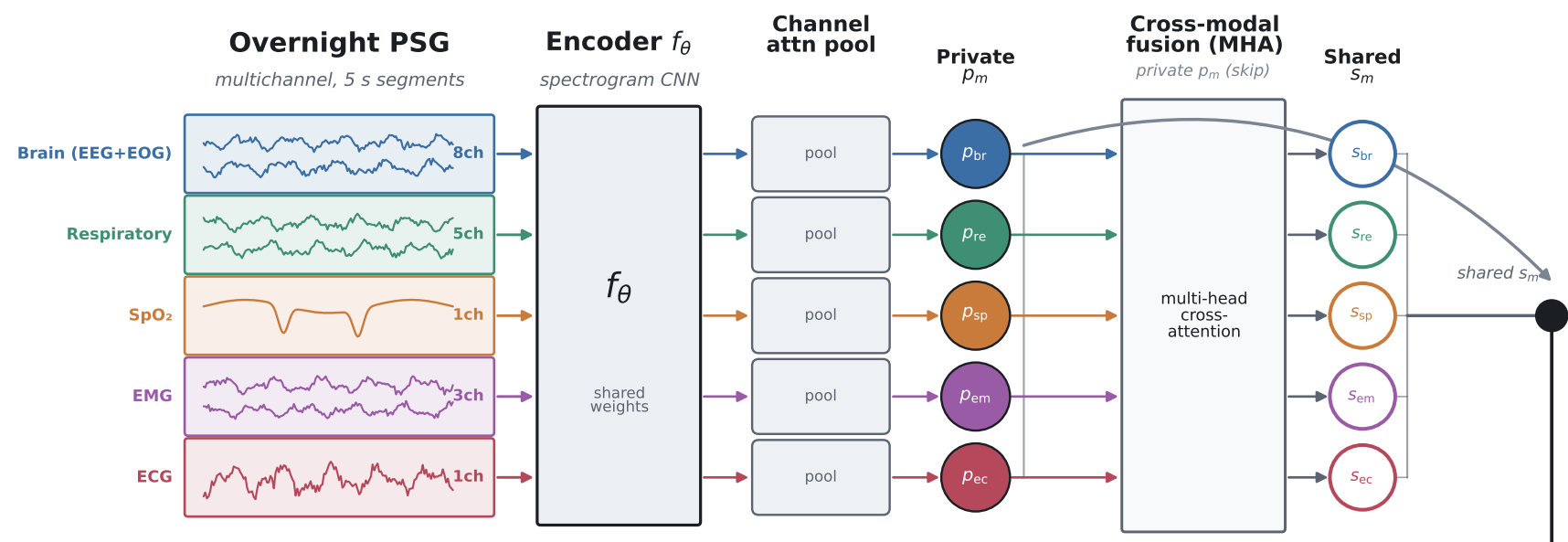
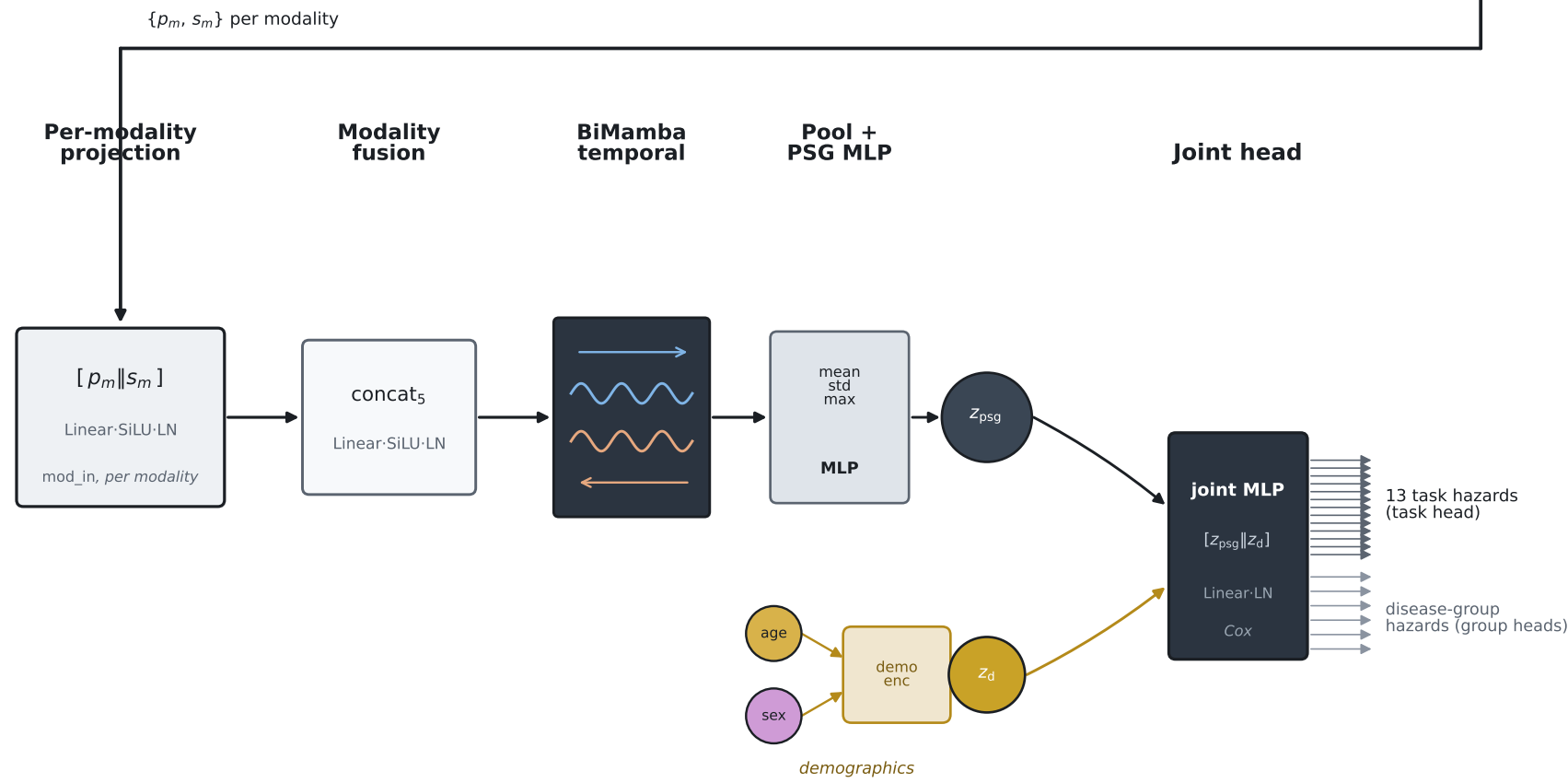


(a) Backbone: per-modality encoding → private → shared



(b) Modality fusion → temporal model → joint demographic head



- Brain (EEG+EOG)
- Respiratory
- SpO₂
- EMG
- ECG